New Phytologist

High-copy sequences reveal distinct evolution of the rye B chromosome

Sonja Klemme[1], Ali Mohammad Banaei-Moghaddam[1], Jiri Macas[2], Thomas Wicker[3], Petr Novák[2] and Andreas Houben[1]

[1]Leibniz Institute of Plant Genetics and Crop Plant Research (IPK), Corrensstraße 3, Gatersleben 06466, Germany; [2]Biology Centre ASCR, Institute of Plant Molecular Biology, Branišovská 31/1160, České Budějovice 370 05, Czech Republic; [3]University of Zurich, Institute of Plant Biology, Zurich 8008, Switzerland

Author for correspondence:
Andreas Houben
Tel: +49 (0) 39482 5486
Email: houben@ipk-gatersleben.de

Received: *18 February 2013*
Accepted: *16 March 2013*

New Phytologist (2013) **199:** 550–558
doi: 10.1111/nph.12289

Key words: B chromosome, genome evolution, late replication, nondisjunction control region, satellite DNA, *Secale cereale* (rye).

Summary

- B chromosomes (Bs) are supernumerary chromosomes that vary in number among individuals of the same species. Because of their dispensable nature, their non-Mendelian inheritance and their origin from A chromosomes (As), one might assume that Bs followed a different evolutionary pathway from As, this being reflected in differences in their high-copy DNA constitution. We provide detailed insight into the composition and distribution of rye (*Secale cereale*) B-located high-copy sequences.
- A- and B-specific high-copy sequences were identified *in silico*. Mobile elements and satellite sequences were verified by fluorescence *in situ* hybridization (FISH). Replication was analyzed via EdU incorporation.
- Although most repeats are similarly distributed along As and Bs, several transposons are either amplified or depleted on the B. An accumulation of B-enriched satellites was found mostly in the nondisjunction control region of the B, which is transcriptionally active and late-replicating. All B-enriched sequences are not unique to the B but are also present in other *Secale* species, suggesting the origin of the B from As of the same genus.
- Our findings highlight the differences between As and Bs. Although Bs originated from As, they have since taken a separate evolutionary pathway.

Introduction

Supernumerary chromosomes that appear in addition to the standard chromosome set exist in many plant, animal and fungal species (Jones & Rees, 1982). These extra chromosomes have been termed B chromosomes (Bs) while the standard chromosomes are called A chromosomes (As). In most cases Bs do not confer any advantage to the host and can even be detrimental if they exceed a certain number. For instance, rye (*Secale cereale*) plants are sterile when they harbor eight Bs (Rees & Ayonoadu, 1973). Bs do not follow Mendelian inheritance; instead, they often accumulate by a 'drive' mechanism (reviewed in Jones, 1991; Jones & Houben, 2003).

Characterization of sequences residing on Bs might shed light on their origin and evolution. Until recently, few sequence data for Bs were available. Early attempts to elucidate the DNA composition of Bs were mainly based on comparative studies of 0B versus +B genomic DNA (Rimpau & Flavell, 1975; Timmis *et al.*, 1975; Sandery *et al.*, 1990; Wilkes *et al.*, 1995). Later, microdissection (Houben *et al.*, 2001a) and flow-sorting (Martis *et al.*, 2012) allowed reliable isolation of B-derived DNA. Bs of many species contain sequences that originated from one or more As (Houben *et al.*, 2001b; Page *et al.*, 2001; Cheng & Lin, 2003; Bugrov *et al.*, 2007; Martis *et al.*, 2012). Even sequences considered as B-specific are also present on As but at low copy number, indicating an intraspecific origin of the B.

As selfish entities, Bs take a distinct path of evolution, and their sequence composition may differ from that of the As. Because Bs are not under selective pressure, mobile elements and other repeats may easily spread and amplify, as in Bs of maize (*Zea mays*) (Lamb *et al.*, 2007), *Brachycome dichromosomatica* (Houben *et al.*, 2001b) and *Plantago* (Dhar *et al.*, 2002). Bs replicating later than As have been shown in the black rat *Rattus rattus* (Raman & Sharma, 1974), the fox *Vulpes fulvus* (Świtoński *et al.*, 1987), the fish *Astyanax scabripinnis* (Maistro *et al.*, 1992) and the frog *Gastrotheca espeletia* (Schmid *et al.*, 2002), although heterochromatic micro-Bs of *B. dichromosomatica* replicate during the entire S-phase (Marschner *et al.*, 2007).

The non-Mendelian accumulation of the rye B by nondisjunction requires factors located at the end of its long arm and at the extended pericentromere (Müntzing, 1948; Håkanson, 1959; Endo *et al.*, 2008; Banaei-Moghaddam *et al.*, 2012). Two B-specific repeat families, E3900 (Blunden *et al.*, 1993) and D1100 (Sandery *et al.*, 1990), reside in the long-arm terminal nondisjunction control region. Both sequences, although heterochromatic, are associated with the euchromatin-specific histone mark H3K4me3 (Carchilan *et al.*, 2007) and transcribe nonconding RNA in anthers, which might be involved in the nondisjunction process.

© 2013 The Authors
New Phytologist © 2013 New Phytologist Trust

Rye Bs of *c.* 560 Mbp (Martis *et al.*, 2012) have been found in accessions from many parts of the world. They are probably of monophyletic origin and similar to those of *Secale segetale* and *Secale ancestrale* (Niwa & Sakamoto, 1995). However, no Bs are known from older rye species such as *Secale strictum* or *Secale sylvestre*. Thus, the origin of the rye B might be linked to the divergence of the *S. cereale* clade from *S. strictum* (Martis *et al.*, 2012).

Employing next-generation sequencing, we determined the DNA composition of flow-sorted rye Bs and As (Martis *et al.*, 2012). The age of rye Bs was estimated to be *c.* 1.1–1.3 million yr. Thus, the rye B originated during or shortly after the radiation of the genus *Secale* (1.7 million yr ago). We proposed that rye Bs are descended from rearrangements of chromosomes 3RS and 7R, with subsequent accumulation of repeats and genic fragments from other A chromosomal regions, as well as insertions of organellar DNA (Martis *et al.*, 2012).

Because of the dispensable nature of the Bs, their non-Mendelian inheritance, and their origin from As, one might assume that the B followed a different evolutionary pathway from the As, this being reflected in differences in their high-copy DNA constitution. Here we provide insight into the composition and distribution of rye B-located high-copy sequences and show that Bs contain a similar proportion of repeats to As, but differ substantially in repeat composition. Although the overall mobile element content is similar, we found a massive accumulation of B-enriched repeats, mostly in the nondisjunction control region at the terminal part of the long arm, which is transcriptionally active and very late replicating, as well as in the extended pericentromere.

Materials and Methods

Plant material and plant cultivation

Plants with Bs from the self-fertile inbred line 7415 of rye (*Secale cereale* L.) (Jimenez *et al.*, 1994) and the related wild rye *Secale strictum* (C. Presl) C. Presl subsp. *strictum* s.l. (Gatersleben Genbank accessions R548) and *Secale sylvestre* Host (Gatersleben Genbank R1116) as well as hexaploid wheat (*Triticum aestivum* L.) with added standard rye Bs (Lindström, 1965) were grown under the same temperature, humidity, and light conditions (16 h : 8 h, light : dark; 22°C day : 16°C night).

In silico identification of B-located repeats

Transposable element (TE)-related sequences were identified by a BLASTN search of individual 454 reads against the TREP repeat database (wheat.pw.usda.gov/ITMI/Repeats/) as previously described (Wicker *et al.*, 2009). In addition, *de novo* identification of repetitive sequences was performed (Martis *et al.*, 2012) using graph-based clustering of 454-sequence reads (accession number EBI-ENA ERP001061). The structure of cluster graphs was investigated using the SeqGrapheR program (Novak *et al.*, 2010). Contig sequences, assembled from reads representing selected parts of cluster graphs, were used to design probes for fluorescence *in situ* hybridization (FISH). These probes were PCR-amplified from genomic DNA, cloned and verified by sequencing. Probe sequences are available from GenBank under accession numbers KC243218–KC243251 for the terminal satellites and KC243252–KC243260 for pericentromeric sequences. The probes corresponding to novel repeats were named based on cluster and contig numbers (e.g. Sc9c11 is *S. cereale* repeat represented by contig 11 from cluster 9).

Fluorescence *in situ* hybridization (FISH)

Probes for FISH were generated by PCR with *Taq* polymerase (Qiagen, Hilden, Germany) from template DNA isolated from plants containing Bs. The annealing temperature was 58°C. The primers used are listed in Supporting Information Table S1. FISH probes were labeled with ChromaTide Texas Red-12-dUTP, Alexa Fluor 488-5-dUTP (http://www.invitrogen.com) or Cy5-dUTP (GE Healthcare Life Sciences; http://www.gelifesciences.com) by nick translation. Mitotic chromosomes were prepared by a method modified after (Kato, 1997): root meristems were washed in ice-cold water and in cold citric buffer (10X citric buffer: 40 ml of 100 mM citric acid and 60 ml of 100 mM tri-sodium citrate), two times each, and then digested in an enzyme cocktail (1% cellulose, 1% pectolyase Y-23, and 1% cytohelicase in citric buffer) for 1 h at 37°C. Afterwards root tips were washed again in citric buffer and in ice-cold ethanol before transfer to an appropriate amount (depending on the number of root tips) of dropping solution (75% acidic acid and 25% methanol). Tissue was disrupted with a dissection needle and 7 µl of cell solution was dropped onto each slide. FISH and Genomic *in situ* hybridization (GISH) were performed as described previously (Gernand *et al.*, 2006; Ma *et al.*, 2010). Chromosomal DNA was denatured at 80°C for 3 min on untreated slides or for 8 min on slides prelabelled with 5-ethynyl-2′-deoxyuridine (EdU). FISH on pachytene chromosomes was performed as described previously (Gonzalez-Garcia *et al.*, 2006). Imaging was performed using an Olympus BX61 microscope and an ORCA-ER CCD camera (Hamamatsu, Hamamatsu City, Shizuoka Pref., Japan). Deconvolution microscopy was employed for superior optical resolution of globular structures. Projections (maximum intensity) were performed with the program AnalySIS (Soft Imaging System, Olympus, Tokyo, Japan). All images were collected in grayscale and pseudocolored with Adobe Photoshop CS5 (Adobe, San Jose, CA, USA).

DNA replication analysis

Roots were treated for 2 h with 15 µM EdU (Click-it EdU Kit; Invitrogen), followed by water for 2.5 h. Fixation was performed with ethanol : acetic acid (3 : 1) for several days. After preparation of chromosomes, a click reaction was performed to detect EdU according to the kit protocol (Baseclick GmbH, Tutzing, Germany). For FISH after replication analysis, slides were washed in 2X SSC (saline-sodium citrate buffer) for 10 min, then dehydrated in 70, 90 and 100% ethanol for 3–5 min each, and

© 2013 The Authors
New Phytologist © 2013 New Phytologist Trust

fixed in ethanol : acetic acid (3 : 1) for at least 1 h. After washing in 2X SSC and fixing with 4% paraformaldehyde for 5 min, FISH was performed as described above (Gernand *et al.*, 2006; Ma *et al.*, 2010).

RT-PCR

For transcription analysis, we performed RT-PCR on cDNA made (RevertAid H Minus first-strand cDNA synthesis kit; Fermentas, Vilnius, Lithuania) from DNase-treated (Ambion TURBO DNase; Invitrogen, Carlsbad, CA, USA) total RNA extracted from the roots, leaves and anthers of plants with and without Bs. The annealing temperature was 58°C. The primers used are listed in Table S1.

Results

Sequencing and bioinformatics data produced previously from flow-sorted As and Bs (Martis *et al.*, 2012) were used to characterize the most abundant repetitive elements of the rye genome. TE-related sequences were identified and classified by similarity search against the TREP repeat database, combined with *de novo* repeat identification using graph-based clustering of sequence reads (Novak *et al.*, 2010). Approximately 90% of the rye genome is repetitive, and 70% is represented by < 60 families of high-copy repeats (Martis *et al.*, 2012). As and Bs revealed similar contents of dispersed repeats but differences regarding sequence and abundance of tandem repeats (Table S2). High-copy repeats were hybridized to selected *Secale* species to determine their chromosomal distribution and evolution. B-specific repeats of the nondisjunction control region were tested for transcriptional activity in different tissues. The known sequences D1100 and E3900 were used for identification of the B (Fig. S1).

With few notable exceptions the distribution of some mobile elements differs between As and Bs

The most abundant mobile elements on the B are Ty3/Gypsy (49%), Ty1/copia (9%) and DNA transposon sequences (4%) (Table S2). The *Copia* elements *Angela* (Fig. 1a) and *Maximus* (Fig. S2A) revealed FISH signals along all chromosome arms, but fewer signals in the pericentromere or subtelomere regions of As and Bs. The *Gypsy* retrotransposons *Fatima* (Fig. S2B) and *WHAM* (wheat abundant mobile DNA; Fig. S2C) are equally distributed on all chromosome arms except at subtelomeres.

Strikingly, the ancient retroelement *Sabrina*, one of the most abundant *Gypsy*-like retroelements in *Triticeae* (Shirasu *et al.*, 2000), is far less abundant on Bs than on As (Fig. 1c). By contrast, the TAR-like *Triticum aestivum* retrotransposon repeat Sc36c82, which belongs to the *Copia* superfamily (Fig. 1d),

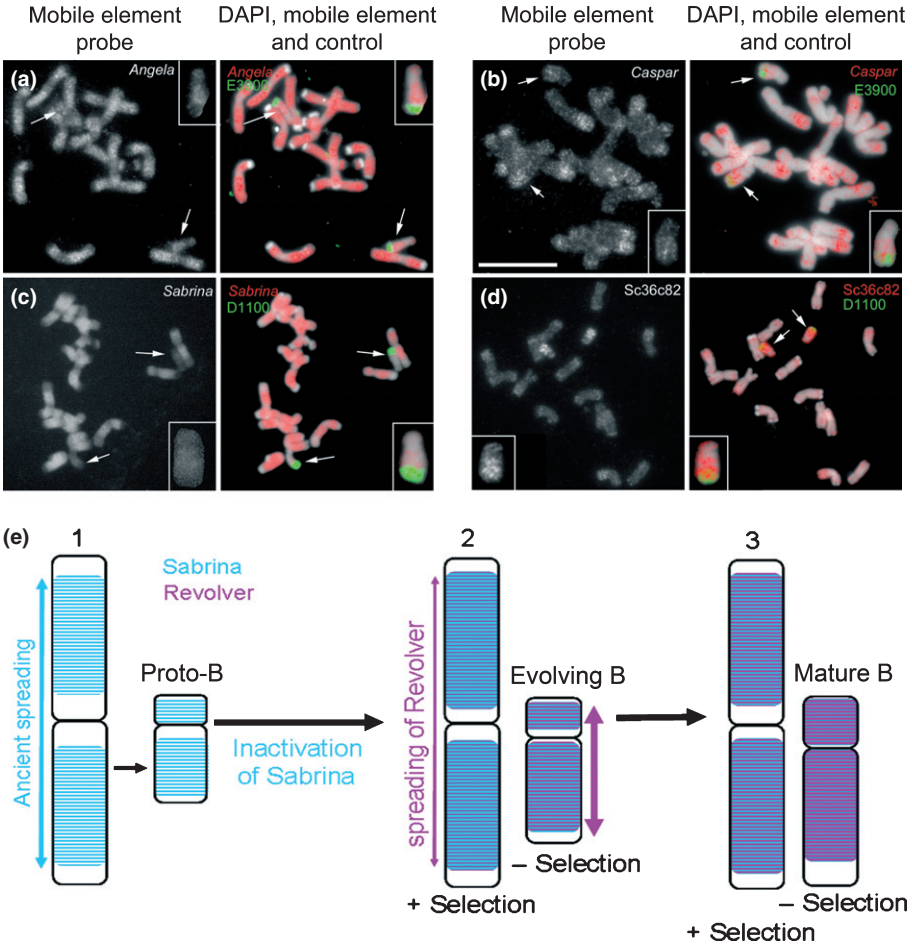


Fig. 1 Mobile element distribution on rye (*Secale cereale*) A and B chromosomes (As and Bs, respectively). Fluorescence *in situ* hybridization (FISH) of mitotic rye +B cells with mobile elements (red) and a B-specific marker (green) was performed. Insets show Bs further enlarged. Bs are arrowed. (a) *Copia* retrotransposon *Angela* and (b) DNA transposon *Caspar* are similarly present on As and Bs. Bar, 10 μm. (c) LTR (long terminal repeat)-retrotransposon *Sabrina* is depleted from Bs while (d) *Copia* predicted sequence Sc36c82 has accumulated. (e) Proposed is the accumulation of the young and active *Revolver* element with simultaneous dilution of the ancient and silent *Sabrina* element. The ancient *Sabrina* element spread through the early *Secale* species and was still present on the B when it was formed. After inactivation of *Sabrina*, *Revolver* spread throughout the genome. Lower selection pressure allowed for more insertion on the B than on As. Thus, *Revolver* accumulates and simultaneously dilutes *Sabrina*.

© 2013 The Authors
New Phytologist © 2013 New Phytologist Trust

displayed on the long arm of the B an intense uniform labeling, while the As revealed less densely dispersed signals, similar to those on chromosomes of *S. strictum* and *S. sylvestre* (data not shown). Sc36c82 is transcriptionally active in roots and anthers and thus possibly still transposing (Fig. 2). The *Gypsy/chromovirus* retrotransposon Sc11c759 (Fig. S2D) is amplified within two euchromatic regions of the B but homogeneously distributed on As. As described by Carchilan *et al.* (2009), the rye-specific *Gypsy*-like retroelement *Revolver* is B-enriched and dispersed across chromosome arms with the exception of subtelomeres (Fig. S2G).

CACTA elements are class II DNA transposons, representing a very abundant super-family in *Triticeae* (Wicker *et al.*, 2003). The members *Caspar* and *Clifford* of the *Caspar* family, the most abundant CACTA elements in rye, are dispersed on As and Bs and additionally accumulate in the heterochromatic subtelomeric regions (Figs 1b, S2E), while the *Caspar*-like element *Jorge* seems more uniformly dispersed (Fig. S2F). The tested CACTA elements showed similar distributions on As and Bs.

In general, we found a similar mobile element distribution between As and Bs, although some elements were either enriched or depleted on the Bs. This hints at a similar origin, but a distinct evolution of the Bs compared with As.

The nondisjunction control region is enriched in B-specific repeats

In addition to the previously described tandem repeats E3900 and D1100 (Sandery *et al.*, 1990; Blunden *et al.*, 1993), we found nine B-enriched sequences yielding clustered FISH signals, indicative of tandem repeats, mostly at the nondisjunction control region of the long B arm and the extended pericentromere (Figs 3, 4, S3) on metaphase as well as on pachytene chromosomes, the latter providing a higher spatial resolution (Fig. S4).

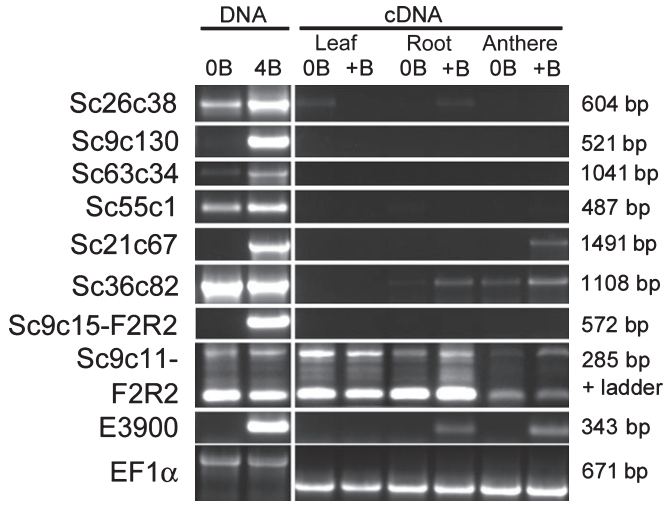


Fig. 2 Transcriptional analysis of rye (*Secale cereale*) B chromosome (B)-enriched sequences. PCR was performed on genomic DNA and cDNA (from leaf, root and anther tissue) of rye plants with and without Bs. Lengths of the achieved fragments are indicated. As a control, the gene elongation factor EF1α and the known B-specific expressed E3900 were included.

Similar to E3900 and D1100, these sequences were characterized by long repeat units spanning up to several kilobases. The bioinformatic similarity-based read clustering was used to identify

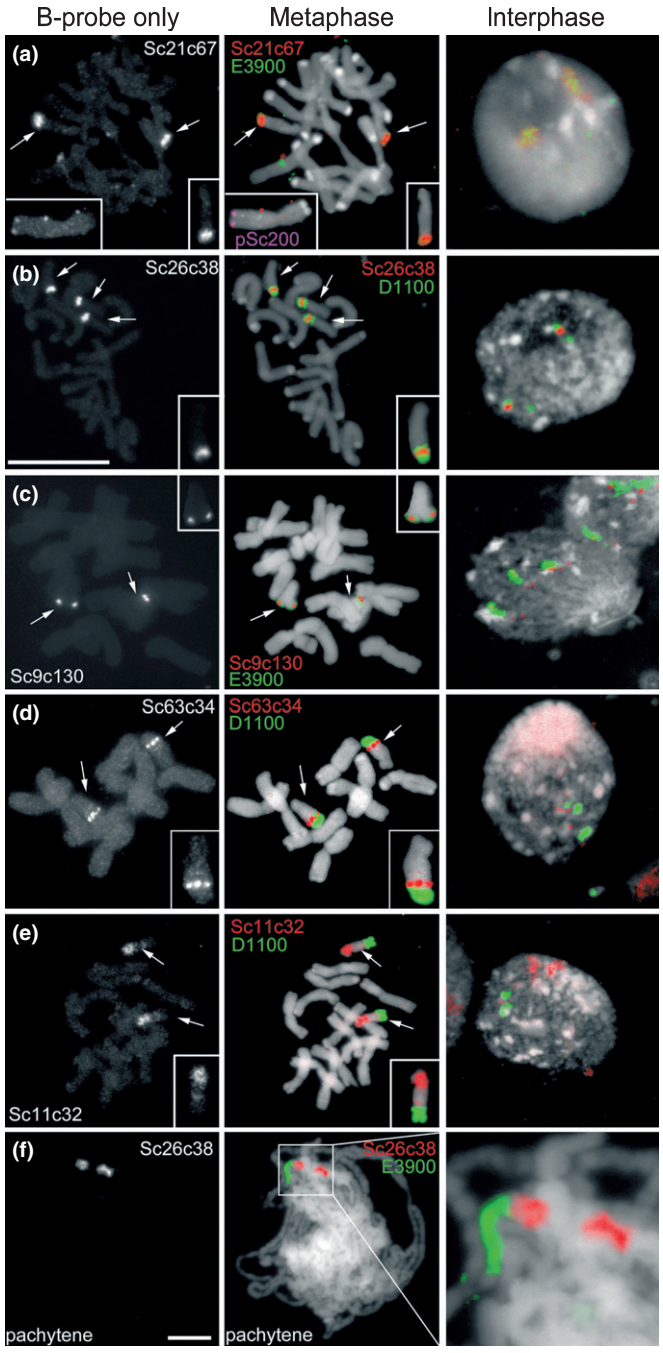


Fig. 3 Chromosomal distribution of rye (*Secale cereale*) B chromosome (B)-enriched tandem repeat sequences. Fluorescence *in situ* hybridization (FISH) of rye +B metaphase and interphase nuclei with tandem repeats (red) and a B-specific marker (green) was performed. Insets show Bs further enlarged. Bs are arrowed. (a) Sc21c67 colocalizes with E3900 and additionally shows signals on one A chromosome pair, where it colocalizes with pSc200 (magenta). (b) Sc26c38 marks a subregion of D1100. (c) Sc9c130 marks the end of the long B arm, (d) Sc63c34 shows an interstitial band and, (e) Sc11c32 marks the pericentromeric area of the B as well as a second region proximal the B-specific marker D1100. (f) Pachytene preparation of Sc26c38 shows two distinct bands. Bars in (b) and (f) 10 μm.

© 2013 The Authors
New Phytologist © 2013 New Phytologist Trust

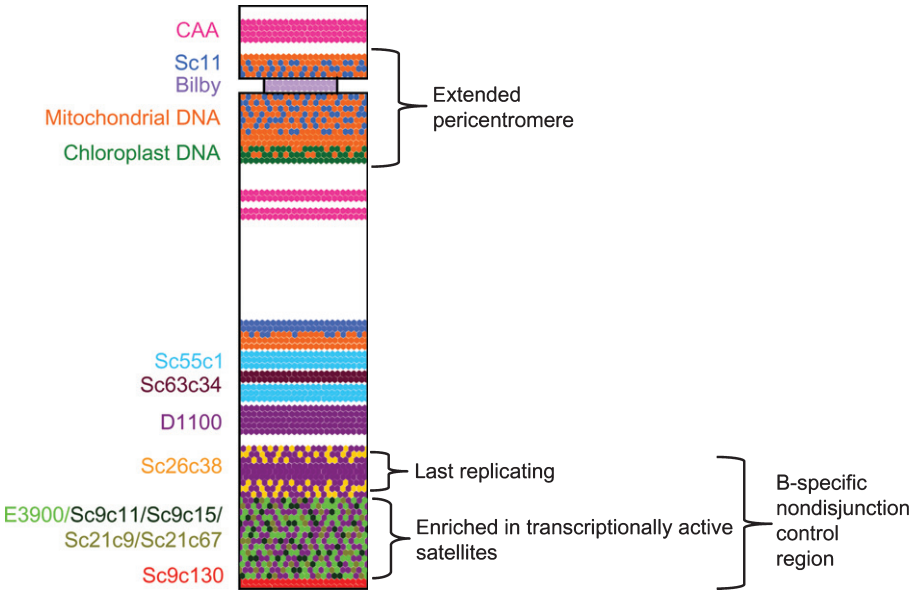


Fig. 4 Model for the distribution of B chromosome (B) marker sequences over the rye (*Secale cereale*) B. The distribution of 14 different B-enriched sequences on the rye B is shown. Each sequence is represented by a color as indicated. Pericentromere, late-replicating, transcriptionally active, and nondisjunction control regions are indicated. The subterminal region of the long arm encompassing the nondisjunction control region is composed of mainly B-specific sequences.

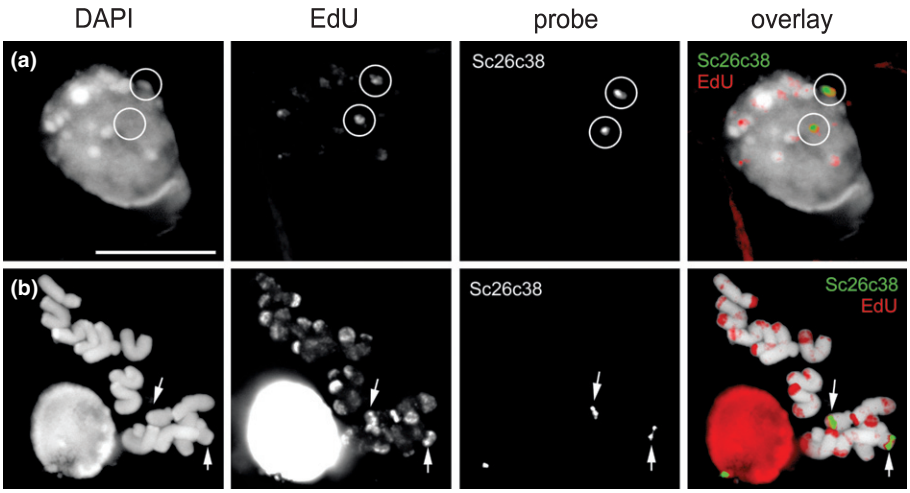


Fig. 5 DNA replication behavior of the rye (*Secale cereale*) B chromosome (B). Replication analysis of mitotic cells of rye +Bs was performed. Late replication of cells was visualized by 5-ethynyl-2′-deoxyuridine (EdU) detection. The position of B-specific probes was detected by subsequent fluorescence *in situ* hybridization (FISH). (a) Sc26c38 colocalizes with EdU labeling late-replicating regions. (b) EdU labeling of metaphases reveals that subterminal repeats replicate late on both As and Bs. Bar, 10 μm.

the repeats. It predicts the higher organization structure such as tandem repeat units and interspersed LTR (long terminal repeat)-retroelement-like sequences (Fig. S5; Movies S1–S3). FISH localization of probes designed from selected regions of the graphs was performed to investigate the spatial arrangement of the potential tandem repeats in distinct chromosome regions.

The graph of the cluster Sc9 includes the repeat D1100 and gag-pol and LTR sequences of *Gypsy* elements (Fig. S5A; Movie S1). The repeat D1100 forms a ring-like structure on the junction of LTR/gag-pol regions, indicating its origin by tandem amplification of a part of the LTR sequence. Another circular structure representing Sc9c130, a putative tandem repeat of 521 bp linked to D1100, is a transcriptionally inactive sequence (Fig. 2) and yielded FISH signals distal of E3900 (Figs 3c, S4D). We also found FISH signals for Sc9c130 on *S. strictum* chromosomes (data not shown). Two additional probes representing linear parts of the graph (Sc9c11 and Sc9c15) labeled the same region as E3900 (Figs S3A,B, S4A). Sc9c15 was not transcribed in leaf, root or anther tissue (Fig. 2). A small region of Sc9c11 (primers F2 + R2; Table S1) was transcribed in all tested tissues with and without Bs.

The second previously identified B-specific repeat, E3900, was found to be a part of cluster Sc21 where it formed a ring-like structure (Fig. S5C; Movie S3). Interestingly, this graph shows the complete sequence of E3900 (Blunden *et al.*, 1993), and a truncated version (Pereira *et al.*, 2009; Fig. S5C, arrowheads). The probes Sc21c9 and Sc21c67, derived from adjacent graph regions but sharing no sequence similarity with E3900, were found to co-localize with E3900. Sc21c67 additionally yielded a very small signal on one pair of As, possibly showing its origin before amplification on the B. The latter signal coincides with the FISH signal for pSc200, which marks exclusively the subterminal heterochromatin on this specific chromosome (Figs 3a, S3C, S4B). Sc21c67 is transcribed only in the anthers of rye plants with Bs (Fig. 2) and undergoes pronounced decondensation during interphase (Fig. 3a). The B-specific transcription makes this sequence an additional candidate for the nondisjunction controlling function.

© 2013 The Authors
New Phytologist © 2013 New Phytologist Trust

The sequence of cluster Sc26c38 localizes within the D1100-positive subregion (Fig. 3b). Condensed metaphase chromosomes display one Sc26c38 signal band. However, on less condensed pachytene chromosomes, two distinct signal bands appear (Figs 3f, S4C). Unlike the decondensed E3900-positive interphase chromatin (Figs 3a,c, S3B), Sc26c38-positive chromatin remains highly condensed during interphase (Fig. 3b). Correlating with its heterochromatic nature, no transcription was detectable (Fig. 2). Interestingly, Sc26c38 is present at chromosome termini of *S. strictum* and near the centromere of one *S. sylvestre* chromosome pair (Fig. S6A).

Sc55c1 revealed weak dispersed signals on Bs, but displayed a FISH pattern on the As (Fig. S3D) that is reminiscent of the highly variable subterminal repeats pSc119.2 and pSc200 of rye (Cuadrado & Jouve, 1994), which are usually absent from the rye B (Tsujimoto & Niwa, 1992; Cuadrado & Jouve, 1994). On the B, Sc55c1 marks two interstitial bands proximal to the D1100 stained region. The distribution of Sc55c1 on *S. strictum* chromosomes is more dispersed.

Sc63c34 shows one band on the long arm of the B chromosome proximal to D1100 and additional faint, dispersed signals along all chromosomes (Figs 3d, S2E). On B it marks the region between the two signal bands of Sc55c1, as visible on the less condensed pachytene chromosomes (Fig. S4E,F). This sequence gives only faint dispersed FISH signals along *S. strictum* chromosomes. Sc9c130- and Sc63c34-positive chromatin does not decondense during interphase (Fig. 3c,d) and does not form transcripts (Fig. 2).

Cluster Sc11 showed the most complex structure, including two rings of reads that were almost exclusively derived from Bs attached to the rest of the graph which contained reads from As and Bs (Fig. S5B; Movie S2). Reads from different parts of the graph showed similarity to gag-pol sequences of either *Gypsy* or *Copia* elements. The part of the graph containing *Gypsy* sequences derived the sequence Sc11c759 described above. The *Copia* sequence Sc11c32 is derived from a junction between the second B-specific ring and the rest of the graph, while Sc11c927 corresponds to the middle of one of the B-specific *Copia* rings. Both sequences were enriched in the B pericentromere adjacent to the core centromere (Figs 3e, S4G,H). A second weak signal band for Sc11c32 appeared in the middle of the long arm. Overexposure revealed weak labeling also on the subtelomeric parts of two A pairs (Fig. S3E), possibly reflecting ancestral *Copia* element positions. Sc11c32 yielded mainly dispersed signals on *S. strictum* and *S. sylvestre* chromosomes (data not shown).

Thus, the nondisjunction control region is composed mainly of B-specifically enriched repeats. Similar repeats occur in lower copy number on As of other *Secale* species, suggesting their specific amplification during the evolution of Bs.

A proximal part of the rye B-nondisjunction control region replicates last in rye and wheat backgrounds

A replication gradient along the rye chromosome arms and late replication of Bs have been suggested previously (Lima-De-Faria & Jaworska, 1972). To compare the DNA replication patterns of As and Bs, EdU, a nucleoside analog of thymidine, was applied during replication. After detection of EdU-labeled DNA, FISH was conducted with Sc26c38 and Sc9c130 probes. The nondisjunction control region of the B replicated last. The remaining B regions exhibit a similar replication behavior to As. Of all sequences tested, Sc26c38 localized closest with the latest replicating areas (Figs 5, S7A). To test whether the replication behavior of the B depends on the host genome, we studied a wheat-rye B addition line. Interphase nuclei displayed again the terminal part of the long B arm as the latest replicating region (Fig. S7B). Multicolor FISH with labeled genomic 0B rye DNA (GISH) and the probe Sc26c38 (Fig. S7C,D) showed that the distal region, including the last-replicating Sc26c38 sequences and encompassing the nondisjunction control region, was composed of mainly B-specific sequences.

Discussion

Mobile elements reflect different intraspecific evolution of As and Bs

Similarity-based clustering of 454-sequence reads efficiently deciphered the high-copy DNA composition and predicted its sequence organization within the rye genome. The similar distribution along As and Bs of most tested mobile sequences such as CACTA elements of the *Caspar* family supports an intraspecific origin of rye Bs.

The ancient retroelement *Sabrina*, abundant in all *Triticeae* and transcriptionally inactive in rye (Shirasu *et al.*, 2000), is less abundant on Bs than on As. The active element *Revolver*, in contrast, as well as the predicted *Copia* transposon Sc36c82 are amplified on the B. The accumulation of active elements might have its cause in the lack of selection pressure on Bs, where the integration of a mobile element does not interrupt functional genes. Meiotic crossing-over has been proposed to remove (hemizygous) mobile elements (Charlesworth *et al.*, 1994). Rye Bs pair frequently with each other and themselves in pachytene (Diez *et al.*, 1993), but bivalents are less connected by chiasmata than the As (Jimenez *et al.*, 2000). Thus, reduced crossing-over might facilitate retroelement accumulation, as proposed for plant Y chromosomes (Charlesworth, 2008). A less likely option is selected transposition to the B, although targeted transposition has been shown for yeast (Zhu *et al.*, 2003) and *Arabidopsis lyrata* (Tsukahara *et al.*, 2012).

A model that explains depletion of *Sabrina* and accumulation of *Revolver* and Sc36c82 is based on different transposition activities of these elements (Fig. 1e). In the *Triticeae* ancestor, *Sabrina* was transposing and spread over the entire genome. After inactivation of *Sabrina* (Shirasu *et al.*, 2000) before or during speciation of rye, the B was formed from the As with *Sabrina* still present. The newly evolving elements such as *Revolver* then became active and transposed throughout the rye genome. The dispensable nature of the B and the lack of selective pressure allowed for stronger accumulation of *Revolver* on the B, even further diluting the remnants of inactive elements which can no longer increase copy number. Less likely is a scenario where the *Sabrina* element transposes predominantly in plants without Bs

© 2013 The Authors
New Phytologist © 2013 New Phytologist Trust

and stays inactive in plants with Bs. Because not all rye plants have Bs, Sabrina should still transpose in 0B rye. Such a behavior has been proposed for the Y chromosome depleted *Ogre* element in *Silene latifolia* (Kejnovsky & Vyskot, 2010).

The *Copia* elements Sc11c32 and Sc11c927, similar to the centromeric sequences *Bilby* and *ScCCS1*, expanded more in the extended pericentromere of the B than in those of As (Banaei-Moghaddam *et al.*, 2012). The presumed ancestral elements are still detectable in subterminal positions on As (Fig. S3E). Also, a B centromere-specific sequence of maize shares homology over 90 bp with the maize knob sequence (Alfenito & Birchler, 1993). Furthermore, in the pericentromere, Sc11 and mitochondrial sequences show a second location near the terminal nondisjunction control region of the rye B (Fig. 3e), indicating an ancient paracentric inversion with breakpoints at the edge of the pericentromere and proximal to the long arm end.

The nondisjunction control region is mainly composed of several B-enriched transcriptionally active tandem repeats

The majority of the newly identified B-enriched tandem repeats map to the nondisjunction control region which is free of signals after genomic *in situ* hybridization with labeled 0B DNA and proximally flanked by the latest replicating sequence Sc26c38 (Fig. S7C,D). Langdon *et al.* (2000) suggested that E3900 and D1100 repeats evolved via amplification of ancestral A-located sequences within the dynamic nondisjunction control region on rye B. The B-enriched tandem repeats could have been amplified via unequal crossover (Smith, 1976).

The maize B long arm end also harbors a region important for nondisjunction (Ward, 1973) which seems to be largely depleted of A sequences (Stark *et al.*, 1996). Four new rye sequences (Sc9c11, Sc9c15, Sc21c9 and Sc21c67) are colocalized in FISH with the B-specific repeat E3900, but not with the D1100, which spans the whole nondisjunction control region (Figs 3a, S3A–C).

Except for Sc9c11, which is transcribed in all three tested tissues in plants with and without Bs, the new tandem repeats displayed transcripts only in +B plants, mostly restricted to anthers where, along with the embryo sacs, post meiotic nondisjunction of rye Bs takes place. Also D1100 and E3900, decondensed during interphase, associated with the euchromatic histone mark H3K4me3 and flanked on either side by strongly heterochromatinized sequences (Sc26c38 and Sc9c130), are transcribed in anthers (Carchilan *et al.*, 2007). Although not yet directly demonstrated, anther-specific transcripts of sequences residing within the terminal nondisjunction control region might be related to the non-Mendelian accumulation of Bs, for example, by mediating stickiness of sister pericentromeres.

The presence of rye B-enriched sequences (e.g. Sc36c82, Sc26c38, Sc9c130, etc.) within the evolutionarily older (De Bustos & Jouve, 2002; Chikmawati *et al.*, 2005; Shang *et al.*, 2006) species *S. strictum* and *S. sylvestre* indicates their amplification after being separated from the standard chromosome complement.

Whether a functional interrelationship exists between the latest replication of Sc26c38 and nondisjunction of rye Bs during first pollen mitosis is not known, but seems unlikely because the stickiness between sister chromatids is restricted to the pericentromeres and the nondisjunction region acts *in trans* (Lima-de-Faria, 1962; Endo *et al.*, 2008).

Although rye Bs possess similar sequence types to As, they vary in their quantitative repeat composition, particularly within the pericentromere and the nondisjunction control region at the terminal part of the long arm. Differences in replication behavior and repeat content highlight on the one hand the origin of Bs from As, and on the other a different evolutionary pathway of As and Bs. The young age of the rye B (*c.* 1.1–1.3 million yr; Martis *et al.*, 2012) makes it a valuable tool with which to study supernumerary chromosome evolution.

Acknowledgements

We thank T. Endo for providing the wheat-rye B addition and K. Pistrick as well as I. Schubert (IPK) for fruitful discussions. We warmly acknowledge the excellent technical assistance of Katrin Kumke, Oda Weiß and Karla Meier and the artistic work of Karin Lipfert. This work was supported by the DFG Germany (HO 1779/10-1/14-1).

References

Alfenito MR, Birchler JA. 1993. Molecular characterization of a maize B chromosome centric sequence. *Genetics* **135**: 589–597.

Banaei-Moghaddam AM, Schubert V, Kumke K, Weiβ O, Klemme S, Nagaki K, Macas J, González-Sánchez M, Heredia V, Gómez-Revilla D *et al.* 2012. Nondisjunction in favor of a chromosome: the mechanism of rye B chromosome drive during pollen mitosis. *Plant Cell* **24**: 4124–4134.

Blunden R, Wilkes TJ, Forster JW, Jimenez MM, Sandery MJ, Karp A, Jones RN. 1993. Identification of the E3900 family, a 2nd family of rye chromosome-B specific repeated sequences. *Genome* **36**: 706–711.

Bugrov AG, Karamysheva TV, Perepelov EA, Elisaphenko EA, Rubtsov DN, Warchalowska-Sliwa E, Tatsuta H, Rubtsov NB. 2007. DNA content of the B chromosomes in grasshopper *Podisma kanoi* Storozh. (Orthoptera, Acrididae). *Chromosome research* **15**: 315–325.

Carchilan M, Delgado M, Ribeiro T, Costa-Nunes P, Caperta A, Morais-Cecilio L, Jones RN, Viegas W, Houben A. 2007. Transcriptionally active heterochromatin in rye B chromosomes. *Plant Cell* **19**: 1738–1749.

Carchilan M, Kumke K, Mikolajewski S, Houben A. 2009. Rye B chromosomes are weakly transcribed and might alter the transcriptional activity of A chromosome sequences. *Chromosoma* **118**: 607–616.

Charlesworth B, Sniegowski P, Stephan W. 1994. The evolutionary dynamics of repetitive DNA in eukaryotes. *Nature* **371**: 215–220.

Charlesworth D. 2008. Plant sex chromosomes. *Genome Dynamics* **4**: 83–94.

Cheng YM, Lin BY. 2003. Cloning and characterization of maize B chromosome sequences derived from microdissection. *Genetics* **164**: 299–310.

Chikmawati T, Skovmand B, Gustafson JP. 2005. Phylogenetic relationships among Secale species revealed by amplified fragment length polymorphisms. *Genome* **48**: 792–801.

Cuadrado A, Jouve N. 1994. Highly repetitive sequences in B chromosomes of *Secale cereale* revealed by fluorescence *in situ* hybridization. *Genome* **37**: 709–712.

De Bustos A, Jouve N. 2002. Phylogenetic relationships of the genus *Secale* based on the characterisation of rDNA ITS sequences. *Plant Systematics and Evolution* **235**: 147–154.

Dhar MK, Friebe B, Koul AK, Gill BS. 2002. Origin of an apparent B chromosome by mutation, chromosome fragmentation and specific DNA sequence amplification. *Chromosoma* **111**: 332–340.

© 2013 The Authors
New Phytologist © 2013 New Phytologist Trust

Diez M, Jiménez MM, Santos JL. 1993. Synaptic patterns of rye B chromosomes. II. The effect of the standard B chromosomes on the pairing of the A set. *TAG Theoretical and Applied Genetics* **87**: 17–21.

Endo TR, Nasuda S, Jones N, Dou Q, Akahori A, Wakimoto M, Tanaka H, Niwa K, Tsujimoto H. 2008. Dissection of rye B chromosomes, and nondisjunction properties of the dissected segments in a common wheat background. *Genes and Genetic Systems* **83**: 23–30.

Gernand D, Rutten T, Pickering R, Houben A. 2006. Elimination of chromosomes in *Hordeum vulgare* x *H. bulbosum* crosses at mitosis and interphase involves micronucleus formation and progressive heterochromatinization. *Cytogenetic and Genome Research* **114**: 169–174.

Gonzalez-Garcia M, Gonzalez-Sanchez M, Puertas MJ. 2006. The high variability of subtelomeric heterochromatin and connections between nonhomologous chromosomes, suggest frequent ectopic recombination in rye meiocytes. *Cytogenetic and Genome Research* **115**: 179–185.

Håkanson A. 1959. Behaviour of different small accessry rye chromosomes at pollen mitosis. *Hereditas* **45**: 623–631.

Houben A, Field BL, Saunders VA. 2001a. Microdissection and chromosome painting of plant B chromosomes. *Methods in Cell Science* **23**: 115–124.

Houben A, Verlin D, Leach CR, Timmis JN. 2001b. The genomic complexity of micro B chromosomes of *Brachycome dichromosomatica*. *Chromosoma* **110**: 451–459.

Jimenez G, Manzanero S, Puertas M. 2000. Relationship between pachytene synapsis, metaphase I associations, and transmission of 2B and 4B chromosomes in rye. *Genome* **43**: 232–239.

Jimenez MM, Romera F, Puertas MJ, Jones RN. 1994. B-Chromosomes in inbred lines of rye (*Secale cereale* L). 1. Vigor and fertility. *Genetica* **92**: 149–154.

Jones N, Houben A. 2003. B chromosomes in plants: escapees from the A chromosome genome? *Trends in Plant Science* **8**: 417–423.

Jones RN. 1991. B-chromosome drive. *American Naturalist* **137**: 430–442.

Jones RN, Rees HH. 1982. *B chromosomes*. London, UK/New York, NY, USA: Academic Press.

Kato A. 1997. An improved method for chromosome counting in maize. *Biotechnic and Histochemistry* **72**: 249–252.

Kejnovsky E, Vyskot B. 2010. Silene latifolia: the classical model to study heteromorphic sex chromosomes. *Cytogenetic and Genome Research* **129**: 250–262.

Lamb JC, Riddle NC, Cheng YM, Theuri J, Birchler JA. 2007. Localization and transcription of a retrotransposon-derived element on the maize B chromosome. *Chromosome Research* **15**: 383–398.

Langdon T, Seago C, Jones RN, Ougham H, Thomas H, Forster JW, Jenkins G. 2000. *De novo* evolution of satellite DNA on the rye B chromosome. *Genetics* **154**: 869–884.

Lima-de-Faria A. 1962. Genetic interaction in rye expressed at chromosome phenotype. *Genetics* **47**: 1455–1462.

Lima-De-Faria A, Jaworska H. 1972. The relation between the chromosome size gradient and the sequence of DNA replication in rye. *Hereditas* **70**: 39–57.

Lindström J. 1965. Transfer to wheat of accessory chromosomes from rye. *Hereditas* **54**: 149–155.

Ma L, Vu GT, Schubert V, Watanabe K, Stein N, Houben A, Schubert I. 2010. Synteny between *Brachypodium distachyon* and *Hordeum vulgare* as revealed by FISH. *Chromosome research* **18**: 841–850.

Maistro EL, Foresti F, Oliveira C, Almeida Toledo LF. 1992. Occurrence of macro B chromosomes in *Astyanax scabripinnis paranae* (Pisces, Characiformes, Characidae). *Genetica* **87**: 101–106.

Marschner S, Kumke K, Houben A. 2007. B chromosomes of B-dichromosomatica show a reduced level of euchromatic histone H3 methylation marks. *Chromosome Research* **15**: 215–222.

Martis MM, Klemme S, Banaei-Moghaddam AM, Blattner FR, Macas J, Schmutzer T, Scholz U, Gundlach H, Wicker T, Šimková H *et al.* 2012. Selfish supernumerary chromosome reveals its origin as a mosaic of host genome and organellar sequences. *Proceedings of the National Academy of Sciences, USA* **109**: 13343–13346.

Müntzing A. 1948. Cytological studies of extra fragment chromosomes in rye. V. A new fragment type arisen by deletion. *Hereditas* **34**: 435–442.

Niwa K, Sakamoto S. 1995. Origin of B-chromosomes in cultivated rye. *Genome* **38**: 307–312.

Novak P, Neumann P, Macas J. 2010. Graph-based clustering and characterization of repetitive sequences in next-generation sequencing data. *BMC Bioinformatics* **11**: 378.

Page BT, Wanous MK, Birchler JA. 2001. Characterization of a maize chromosome 4 centromeric sequence: evidence for an evolutionary relationship with the B chromosome centromere. *Genetics* **159**: 291–302.

Pereira HS, Barao A, Caperta A, Rocha J, Viegas W, Delgado M. 2009. Rye Bs disclose ancestral sequences in cereal genomes with a potential role in gametophyte chromatid segregation. *Molecular Biology and Evolution* **26**: 1683–1697.

Raman R, Sharma T. 1974. DNA replication, G- and C-bands and meiotic behaviour of supernumerary chromosomes of *Rattus rattus* (Linn.). *Chromosoma* **45**: 111–119.

Rees H, Ayonoadu U. 1973. B-chromosome selection in rye. *Theoretical and Applied Genetics* **43**: 162–166.

Rimpau J, Flavell RB. 1975. Characterization of rye B chromosome DNA by DNA-DNA hybridization. *Chromosoma* **52**: 207–217.

Sandery MJ, Forster JW, Blunden R, Jones RN. 1990. Identification of a family of repeated sequences on the rye B-chromosome. *Genome* **33**: 908–913.

Schmid M, Ziegler CG, Steinlein C, Nanda I, Haaf T. 2002. Chromosome banding in Amphibia: XXIV. The B chromosomes of *Gastrotheca espeletia* (Anura, Hylidae). *Cytogenetic and Genome Research* **97**: 205–218.

Shang H-Y, Wei Y-M, Wang X-R, Zheng Y-L. 2006. Genetic diversity and phylogenetic relationships in the rye genus *Secale* L. (rye) based on *Secale cereale* microsatellite markers. *Genetics and Molecular Biology* **29**: 685–691.

Shirasu K, Schulman AH, Lahaye T, Schulze-Lefert P. 2000. A contiguous 66-kb barley DNA sequence provides evidence for reversible genome expansion. *Genome Research* **10**: 908–915.

Smith GP. 1976. Evolution of repeated DNA sequences by unequal crossover. *Science* **191**: 528–535.

Stark EA, Connerton I, Bennett ST, Barnes SR, Parker JS, Forster JW. 1996. Molecular analysis of the structure of the maize B-chromosome. *Chromosome Res* **4**: 15–23.

Świtoński M, Gustavsson I, Höjer K, Plöen L. 1987. Synaptonemal complex analysis of the B-chromosomes in spermatocytes of the silver fox (*Vulpes fulvus* Desm.). *Cytogenetic and Genome Research* **45**: 84–92.

Timmis JN, Ingle J, Sinclair J, Jones N. 1975. The genomic quality of rye B chromosomes. *Journal of Experimental Botany* **26**: 367–378.

Tsujimoto H, Niwa K. 1992. DNA structure of the B-chromosome of rye revealed by *in situ* hybridization using repetitive sequences. *Japanese Journal of Genetics* **67**: 233–241.

Tsukahara S, Kawabe A, Kobayashi A, Ito T, Aizu T, Shin-i T, Toyoda A, Fujiyama A, Tarutani Y, Kakutani T. 2012. Centromere-targeted *de novo* integrations of an LTR retrotransposon of *Arabidopsis lyrata*. *Genes & development* **26**: 705–713.

Ward EJ. 1973. Nondisjunction: localization of the controlling site in the maize B chromosome. *Genetics* **73**: 387–391.

Wicker T, Guyot R, Yahiaoui N, Keller B. 2003. CACTA transposons in Triticeae. A diverse family of high-copy repetitive elements. *Plant Physiology* **132**: 52–63.

Wicker T, Taudien S, Houben A, Keller B, Graner A, Platzer M, Stein N. 2009. A whole-genome snapshot of 454 sequences exposes the composition of the barley genome and provides evidence for parallel evolution of genome size in wheat and barley. *Plant Journal* **59**: 712–722.

Wilkes TM, Francki MG, Langridge P, Karp A, Jones RN, Forster JW. 1995. Analysis of rye B-chromosome structure using fluorescence *in-situ* hybridization (Fish). *Chromosome Research* **3**: 466–472.

Zhu Y, Dai J, Fuerst PG, Voytas DF. 2003. Controlling integration specificity of a yeast retrotransposon. *Proceedings of the National Academy of Sciences, USA* **100**: 5891–5895.

Supporting Information

Additional supporting information may be found in the online version of this article.

© 2013 The Authors
New Phytologist © 2013 New Phytologist Trust

Fig. S1 Established controls for the rye B.

Fig. S2 Mobile element distribution on rye A and B chromosomes.

Fig. S3 Chromosomal distribution of rye B-enriched tandem repeat sequences.

Fig. S4 B-enriched marker sequences visualized on pachytene chromosomes.

Fig. S5 Visualization of clusters Sc9, Sc11 and Sc21.

Fig. S6 B-enriched sequences are present in evolutionarily older *Secale* species.

Fig. S7 Replication analysis of mitotic cells of rye +Bs.

Table S1 Primers used for RT-PCR and for the generation of FISH probes

Table S2 Repeat proportions as detected in the different sequencing data sets

Movie S1 Rotation animation of the 3D visualization graph of cluster Sc9 (important sequences on the graphs have been highlighted in color; see also Fig. S5A).

Movie S2 Rotation animation of the 3D visualization graph of cluster Sc11 (important sequences on the graphs have been highlighted in color; see also Fig. S5B.)

Movie S3 Rotation animation of the 3D visualization graph of cluster Sc21 (important sequences on the graphs have been highlighted in color; see also Fig. S5C.)

Please note: Wiley-Blackwell are not responsible for the content or functionality of any supporting information supplied by the authors. Any queries (other than missing material) should be directed to the *New Phytologist* Central Office.



About *New Phytologist*

- *New Phytologist* is an electronic (online-only) journal owned by the New Phytologist Trust, a **not-for-profit organization** dedicated to the promotion of plant science, facilitating projects from symposia to free access for our Tansley reviews.
- Regular papers, Letters, Research reviews, Rapid reports and both Modelling/Theory and Methods papers are encouraged. We are committed to rapid processing, from online submission through to publication 'as ready' via *Early View* – our average time to decision is <25 days. There are **no page or colour charges** and a PDF version will be provided for each article.
- The journal is available online at Wiley Online Library. Visit **www.newphytologist.com** to search the articles and register for table of contents email alerts.
- If you have any questions, do get in touch with Central Office (np-centraloffice@lancaster.ac.uk) or, if it is more convenient, our USA Office (np-usaoffice@ornl.gov)
- For submission instructions, subscription and all the latest information visit **www.newphytologist.com**

© 2013 The Authors
New Phytologist © 2013 New Phytologist Trust